

Divisive Clustering - Explanation with Working Example

Divisive Clustering is a hierarchical clustering technique used in unsupervised machine learning. It follows a top-down approach, where all data points start in one large cluster and are recursively split into smaller clusters until the desired clustering structure is achieved.

How Divisive Clustering Works

1. Start with all data points in a single cluster.
2. Find the cluster with the highest dissimilarity.
3. Split the cluster into smaller clusters.
4. Repeat the splitting process recursively.
5. Continue until the required number of clusters is formed.

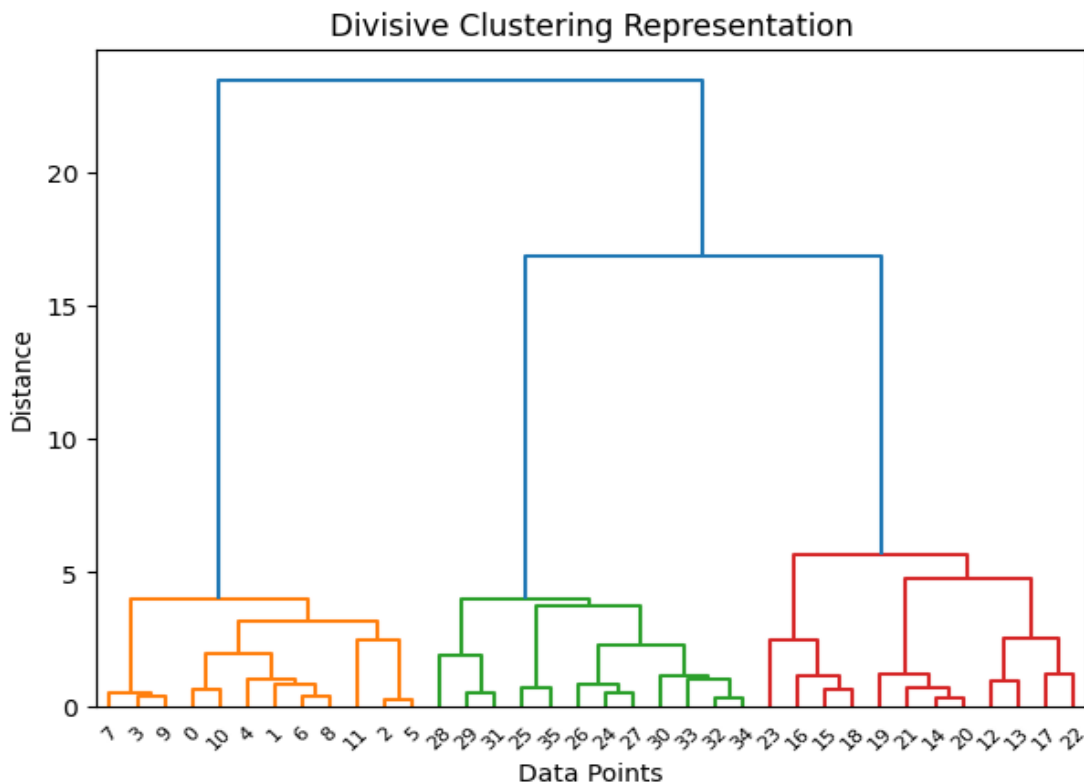
Divisive vs Agglomerative Clustering

Divisive Clustering: Top-down approach (split clusters). **Agglomerative Clustering:** Bottom-up approach (merge clusters). Divisive clustering is generally more computationally expensive but can produce meaningful cluster structures.

Working Example

Suppose we have customer purchasing data for an online store.

Initially: All customers belong to one large cluster. The algorithm identifies the most dissimilar customers. The cluster is divided into smaller groups. The process continues until meaningful customer segments are formed. This helps businesses identify different customer categories and behaviors.



Advantages of Divisive Clustering

Provides a clear hierarchical cluster structure. Can produce accurate clustering for certain datasets. Useful for understanding large data relationships.

Limitations of Divisive Clustering

Computationally expensive for large datasets. More complex than K-Means and Agglomerative clustering. Requires careful selection of splitting criteria.

Conclusion

Divisive Clustering is a hierarchical clustering method that starts with one large cluster and divides it into smaller clusters recursively. It is useful for discovering hidden structures and relationships within data.